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REVIEW

A review of predator–prey systems with dormancy of predators

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Abstract The predator–prey system has received much attention in the field of ecology and evolution. The interaction and competition between populations in nature can be described by the predator–prey system. Under large-amplitude fluctuations caused by harsh environmental conditions, the dormant progeny has been found as an effective strategy to prevent extinction. In this review paper, recent developments of dormancy in predator–prey systems are reviewed. The significant impacts of dormancy on the competition and evolution in predator–prey systems are then discussed through different models. The connections between dormancy in predator–prey systems and the game-theoretic Parrondo's paradox are also discussed: the dormitive predator with inferior traits can outcompete the perennially active predator by switching between two losing strategies. Future outlook about the dormancy research in predator–prey systems is also discussed.

Keywords Predator–prey system · Dormancy · Evolutionary dynamics · Nonlinear analysis · Parrondo's paradox · Game theory

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1 Introduction

The predator–prey system has been identified as the building block of biological systems because it can describe the growth, interaction, competition, and coevolution between different species [1–5]. All species need to seek sufficient resources in the environment to sustain their existence and prevent extinction despite taking different forms [6–11], such as parasite-host, tumor cells-immune system, plant-herbivore, and susceptible-infectious interactions. Due to the complicated relationship between different populations and varying environments, population size is usually not constant but changes significantly over time. Such population fluctuations in the environment have been found to be limited by resources, including food and nest [12, 13]. In addition, there are two forms of control that can affect the population size and population cycles over a period of time [14–16]: (1) bottom-up control due to the competition under limited resources; (2) top-down control due to predation. Therefore, each individual needs to obtain sufficient resources and prevent becoming food for predators at the same time to survive and

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grow in the environment. The microcosm experiments about predator–prey systems have shown the oscillatory time series with unprecedented length. In the system, irregular and non-coherent oscillations will return to regular and coherent oscillations with a phase relationship due to stochasticity, thereby demonstrating the infinite persistence of populations [17–19]. Therefore, it is important to combine experiments with theoretical mathematical models to study the role of transient dynamics in the real-world [20]. The classical example for the predator–prey system is the interaction between snowshoe hare and lynx [21,22], and researchers have developed numerous mathematical models to describe the relationship of species, including differential equation models. Specifically, the Lotka–Volterra model [23–26] is a pair of first-order nonlinear differential equations and is arguably the simplest model in mathematical ecology. The Rosenzweig–MacArthur model [27,28] was then extended by using functional response [29–35] and density-based growth model [36] to uncover real-world conditions [37–45] . There is also the Arditi–Ginzburg Model [46] that is based on the ratio-dependent predator–prey dynamics to explain heterogeneity in biological systems. These models not only can be used to describe the population dynamics of predator and prey [47–50], but also applicable across many different fields such as epidemics [51–56], ecology [57–62], and cryptocurrencies [63,64].

In order to reduce the risk of extinction, predators always produce dormant progeny (resting eggs) and hibernate in the population cycle, which has been widely found in zooplankton and insects [65–70]. Therefore, dormancy has been introduced as an effective strategy to render species to survive under harsh environmental conditions [71–77] . However, dormitive predators are different from perennially active predators because (1) only a portion of energy is used to produce active progeny; and (2) the remaining energy is used to produce dormant offspring. In addition, the dormant form cannot grow on its own and needs extra energy to produce the hardy resting egg, such as zooplankton [78,79]. How does dormancy exist as a competitive behavior in the environment even though it is much costly compared with keeping active? There have been many studies about dormancy in the predator–prey system, for example, analyzing the stability and dynamic behavior [80,81], investigating the bifurcation structure and chaos [82–87] , and uncovering the impact of parameters [78]. These studies have shown that dormancy can reduce the extinction risk by suppressing the emergence of population fluctuations, thereby accounting for the nonoccurrence of the paradox of enrichment in predator–prey systems [88,89]. In general, the induction of dormancy is determined by the density of prey through sigmoid switching function, but predators can also enter dormancy based on the variation of prey density [90] which can stabilize the dynamics of the system. In addition, impulsive perturbations [91,92] have been introduced in predator–prey systems to study its impact on the inherent oscillation which can be further used in ecosystem management. Turing instabilities in the predator–prey reaction-diffusion systems [93] have also been applied to analyze the effect of dormancy on spatio-temporal patterns.

There are only prey and dormitive predator in these studies, which is useful to analyze the effect of dormancy on predator–prey systems. However, it is insufficient to explain the evolutionary stability and competitive advantage of dormancy. In addition, dormitive predators usually coexist and compete with the perennially active predator in the environment. Therefore, Tan et al. proposed a population model to study how dormitive predator and perennially active predator compete over the same prey population [94]. They found that the active form and dormant form of dormitive predators are both losing strategies compared with the perennially active predator because the active form needs to consume extra energy to produce dormant progeny and the dormant form cannot grow on its own. However, the dormitive predator with inferior traits can outcompete the perennially active predator by switching between two losing strategies, which can be explained by the game-theoretic Parrondo's paradox [95–100] , an abstraction of the phenomenon of flashing Brownian ratchets [101–104]. In addition, they found that the dormitive predator is more advantageous under a higher level of resource enrichment, which appears counterintuitive [94]. Parrondo's paradox has also been used to explain how disadvantaged species can achieve a winning outcome by switching between losing strategies in biological and ecological context, including organisms [105–108], epidemiology [109], territorial expansion [110], and temperate bacteriophages [111]. The competition between the perennially active prey and dormitive prey has also been studied when both of them are preyed on by the same predator population [112]. They found that the disadvantaged dormitive prey can prevent extinction by entering dormancy

and win the competition through switching between different forms, which can be explained by Parrondo's paradox.

The rest of this paper is organized as follows. The classical predator–prey system (Lotka–Volterra model) that describes the relationship between foxes and rabbits is introduced in Sect. 2. Due to the importance of dormancy in predator–prey systems, it is discussed in detail for several different systems in Sect. 3, including a consumer-resource system, a reaction-diffusion system, and a fast-slow system. In order to investigate how dormitive species can outcompete the perennially active species by switching between losing strategies, there are two subsections in Sect. 4, including predator dormancy in Sect. 4.1 and prey dormancy in Sect. 4.2 . The conclusions and future outlook in this field are discussed in Sect. 5. All figures shown in this review paper are obtained by using MATLAB's ode23 function (an implementation of an explicit Runge–Kutta (2,3) pair of Bogacki and Shampine). The absolute error tolerance and relative error tolerance are set as 10^{-8} in the simulations.

2 Predator–prey system

In the classical predator–prey system, the relationships between the predator (foxes, z) and prey (rabbits, p) are modeled by two coupled nonlinear differential equations.

The assumptions of this model are:

(1) The resources (clover) supplied for the growth of rabbits are unlimited, thus causing the exponential reproduction of prey with rate r. Rabbits are the only food source for foxes;
(2) In the absence of interaction, the rate of change of the population will only be related to itself. This means that rabbits will grow exponentially without foxes and foxes can only die without rabbits;
(3) The predation rate is determined by the frequency of encounters between rabbits and foxes, that is, the product of p and z.

In this model, the predator will capture and consume prey when they meet with rate α. However, the growth rate of z is not necessarily the same as the consumption rate, so the growth rate is $\beta \leq \alpha$. The mortality rate of predator is d_z, which is close to the exponential decay without prey. These assumptions are captured by the differential equations

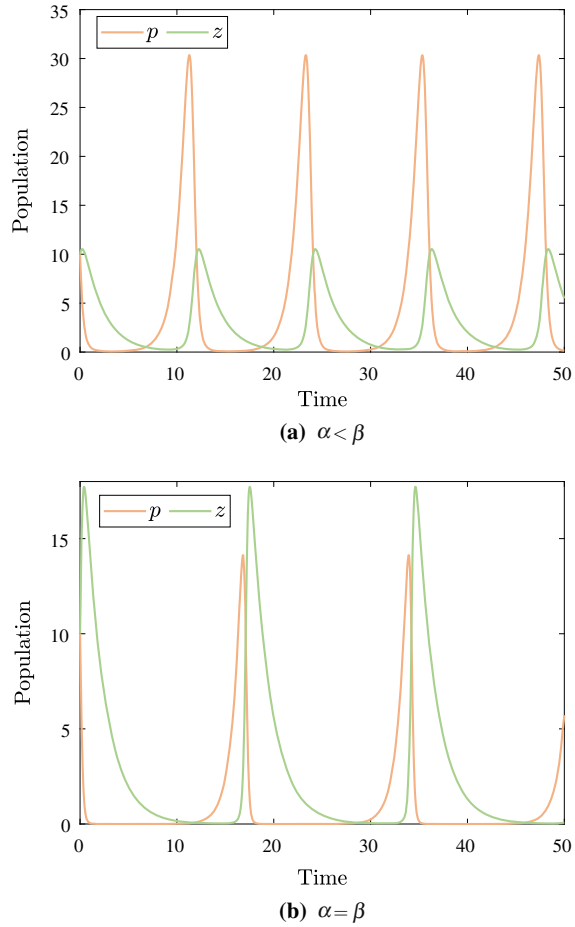


Fig. 1 Dynamics of the predator–prey system under different conditions. The parameter values are $r = 1.1$, $\alpha = 0.4$, and $d_z = 0.5$. (**a**) $\alpha > \beta = 0.1$. (**b**) $\alpha = \beta = 0.4$

$$\begin{aligned} \frac{dp}{dt} &= rp - \alpha pz, \\ \frac{dz}{dt} &= \beta pz - d_z z. \end{aligned} \tag{1}$$

When the initial density of two populations is set as $p_0 = 10, z_0 = 10$, the results are shown in Fig. 1. The peak density of z is always lower than the peak density of p when $\alpha < \beta$ (Fig. 1a), but the opposite happens when $\alpha = \beta$ (Fig. 1b) because predators can obtain more energy with the same number of prey. The near extinction of each species leads to the growth of the other species, but both populations will not become extinct in population cycles.

3 Dormitive predator in predator–prey systems

In the environment, several species have developed adaptive strategies to prevent extinction under harsh environmental conditions, such as hibernation in mammals [113–115], using seed banks in plant ecology [116,117], and the diapause of insects and zooplankton [66]. Therefore, dormancy has become a common and natural response to stress. Kuwamura et al. [89] have proposed a mathematical model of a predator–prey system where the predator can produce resting eggs under harsh environmental conditions

$$\begin{aligned} \frac{dp}{dt} &= r\left(1-\frac{p}{K}\right)p - f(p)z_1, \\ \frac{dz_1}{dt} &= k_1\mu(p)f(p)z_1 + \alpha z_2 - d_1 z_1, \\ \frac{dz_2}{dt} &= k_2(1-\mu(p))f(p)z_1 - \alpha z_2 - d_2 z_2, \end{aligned} \tag{2}$$

where p is the density of prey, z_1 and z_2 represent the density of the active form and the dormant form of predators. In this model, the resource supplied for the growth of prey is limited, thus it follows the logistic growth model $r(1-p/K)$ where r is the intrinsic growth rate and K is the environment carrying capacity. The predation rate $f(p)$ is a monotone increasing function based on the density of p, which follows the Holling type II functional response [118],

$$f(p) = \frac{cp}{1+chp}, \tag{3}$$

where c and h are the foraging efficiency and handling time, respectively. The predation function $f(p)$ is close to cp as a linear function when p is small and converges to $1/h$ when p approaches infinity. Apart from Eq. (3), the predation rate $f(p)$ [84] can be also represented by

$$f(p) = \frac{cp}{b+p}, \tag{4}$$

where c and b are the foraging efficiency and the half saturation constant, respectively.

The growth rates of active predator z_1 and dormant predator z_2 are related to the increasing rates $k_1 = e/m_1$ and $k_2 = e/m_2$, where m_1 and m_2 are the energy needed to produce the active and dormant progeny, respectively. The efficiency of converting food into energy is represented by e. The energy cannot be fully used to produce the active offspring. Only $\mu(p)$ is used to produce the active form, and the remaining energy $(1-\mu(p))$ is given to the dormant offspring z_2. Two types of predator die with the mortality rate d_1 and d_2, respectively. The active form z_1 can be also supplemented by dormant form z_2 with hatching rate α, which means the average dormancy period is $1/\alpha$.

The transition function is

$$\mu(p) = S\left(\frac{g(p)-\eta}{\sigma}\right), \tag{5}$$

where η and σ represent the switching threshold and the shape of the switching function, respectively. S is a monotonically increasing bounded function shown as follows,

(1) $0 < S < 1$;
(2) $S(-\infty) = 0$;
(3) $S(+\infty) = 1$.

The sigmoid function [119,120] is usually chosen as the switching function $\mu(p)$,

$$\mu(p) = S(p) = \frac{1}{1+e^{-p}} = \frac{1+\tanh p}{2}. \tag{6}$$

It is worth noting that the system in Eq. (2) will degenerate to the classical predator–prey system (the MacArthur-Rosenzweig model [27]) when the switching function $\mu(p) \equiv 1$ and hatching rate $\alpha \equiv 0$,

$$\begin{aligned} \frac{dp}{dt} &= r\left(1-\frac{p}{K}\right)p - f(p)z_1, \\ \frac{dz_1}{dt} &= k_1 f(p)z_1 - d_1 z_1, \end{aligned} \tag{7}$$

where prey follows the logistic growth model here.

In general, we assume that (1) the energy needed to produce resting eggs is larger than the energy needed to produce subitaneous eggs (active offspring), and (2) the mortality rate of dormant form is less than the active form, that is,

$$\begin{aligned} &\text{Energy consumption}: k_1 \geq k_2; \\ &\text{Mortality rate}: d_1 > d_2. \end{aligned} \tag{8}$$

Two equilibria for Eq. (2) except for the trivial equilibrium (0, 0, 0) are shown as:

$$(p, z_1, z_2) = (K, 0, 0) \text{ and } (p^*, z_1^*, z_2^*), \tag{9}$$

which can be obtained by,

$$\begin{aligned} d_1 &= k_1\mu(p^*)f(p^*) + \frac{k_2\alpha}{\alpha+d_2}v(p^*)f(p^*), \\ z_1^* &= \frac{r(1-p^*/K)p^*}{f(p^*)}, \\ z_2^* &= \frac{k_2}{\alpha+d_2}(1-\mu(p^*))f(p^*)z_1^*. \end{aligned} \tag{10}$$

It is obvious that p^* is not related to K, but z_1^* and z_2^* depend on K. In addition, $p^* = p^*(\alpha)$ is monotonously decreasing with α. It has been proven that (p^*, z_1^*, z_2^*) can bifurcate transcritically from $(K, 0, 0)$ with the increase of K [89]. More descriptions about two equilibria are shown below,

(1) The first equilibrium $(K, 0, 0)$ is only stable when $0 < K < K^{tc}$ but unstable when $K > K^{tc}$, where K^{tc} is called the transcritical bifurcation point and follows the condition $K^{tc} = K^{tc}(\alpha) = p^*(\alpha)$. It can be also observed in the MacArthur–Rosenzweig model (Eq. 7).
(2) Furthermore, the second equilibrium (p^*, z_1^*, z_2^*) is stable when $K^{tc} < K < K^h$ but unstable when $K > K^h$, where K^h is called the Hopf bifurcation point. All populations will coexist in the system (Eq. 2).
(3) When $\alpha = 0$, the transcritical bifurcation point K^{tc} and the Hopf bifurcation point K^h are defined as,

$$K^{tc} = p^*, \quad K^h = \frac{2p^* f(p^*) - f'(p^*)(p^*)^2}{f(p^*) - f'(p^*)p^*}. \tag{11}$$

The proof and derivation about K^{tc} and K^h can be found in Refs. [84,89]. The bifurcation structure of this model supports the existence of the large amplitude cycle and stable equilibrium in the resource-rich environment. In addition, predator dormancy can cause the stability of the predator–prey system with sharp switching function (Eq. 6) or small values of α and d_2 [121].

In order to analyze the effect of η on the system, we consider the system (Eq. 2) with $f(p)$ shown in Eq. (3) and the switching function:

$$\mu(p) = \frac{1}{2}\left(\tanh\left(\frac{p-\eta}{\sigma}\right) + 1\right) = \frac{1}{1 + \exp\left(\frac{-2(p-\eta)}{\sigma}\right)}, \tag{12}$$

for the parameter setting:

$$c = 0.4, h = 0.75, r = 1, \sigma = 0.2, K = 40, \alpha = 0.2, k_1 = 1, k_2 = 0.5, d_1 = 1, d_2 = 0.05. \tag{13}$$

The mortality rate d_2 of dormant form is suggested to be very small compared with d_1 [122], thus the decline of z_2 is mainly caused by hatching. The interaction in the system under different values of η is shown in Fig. 2. With large negative η, almost all predator progeny become active form and there is no dormant predator in the system (Fig. 2a). With increasing η, all populations will coexist in the environment (Fig. 2b). In this situation, the density of prey cannot reach the environmental carrying capacity because there are still active predator to forage prey to keep its density lower than the carrying capacity. When η is higher and equals 20, the system becomes stable and the density of each population will not change over time (Fig. 2c).

Under the parameter condition from previous works [123–126] shown below,

$$c = 1, h = 0.5, r = 1, \eta = 0.65, \sigma = 0.1, k_1 = 0.5, k_2 = 0.1, d_1 = 0.2, d_2 = 0.05, \tag{14}$$

the phase portraits with different values of α and K are shown in Fig. 3. The density fluctuations of each population will exist in the system and increase with higher values of K. With the increasing of K, there will be an attracting mixed-mode oscillatory solution, and other solutions of the system (Eq. 2) will converge to it. In addition, this mixed-mode oscillation will suffer a series of bifurcations and there will be a chaotic attractor [84]. The hatching rate α does not have a clear impact on the density fluctuation (Fig. 3).

Dormancy of predators not only can be determined by the density of prey, but also by the rate of descent in the prey density [90], which can allow the predator-prey system to be stabilized. Different from the system in Eq. (2) where $\mu(p)$ (Eq. 12) is determined by current density of prey, the switching function is shown as,

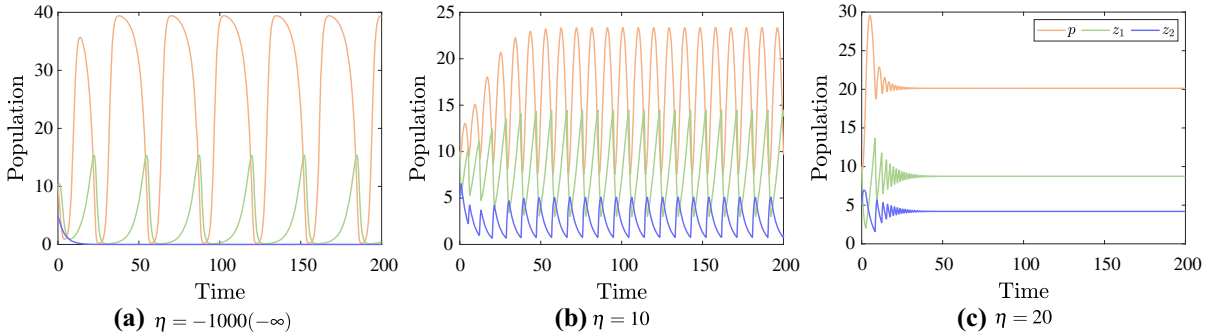
$$\mu(\dot{p}) = \frac{1}{2}(\tanh(\sigma(\dot{p} - \eta)) + 1) = \frac{1}{1 + \exp(-2\sigma(\dot{p} - \eta))}, \tag{15}$$

where $\dot{p}=dp/dt$ is the derivative of prey density p, and $\sigma = 2\mu'(0)$ here. In this situation, there are also two equilibria with following properties:

(1) $(K, 0, 0)$ is stable for $0 < K < K^{tc}$ but unstable for $K > K^{tc}$, where K^{tc} equals p^*.
(2) (p^*, z_1^*, z_2^*) is shown as,

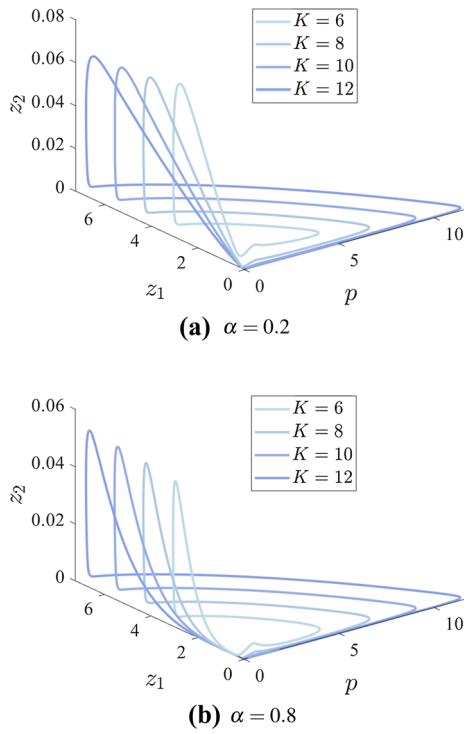
$$d_1 = k_1\mu_0 f(p^*) + \frac{k_2\alpha}{\alpha + d_2}(1 - \mu_0) f(p^*), \quad z_1^* = \frac{r(1 - p^*/K)p^*}{f(p^*)}, \quad z_2^* = \frac{k_2}{\alpha + d_2}(1 - \mu_0) f(p^*) z_1^*, \tag{16}$$

where $\mu_0 = \mu(0)$. $A = (a_{ij})$ is the linearized matrix of the right side of the system at



(a) $\eta = -1000(-\infty)$ **(b)** $\eta = 10$ **(c)** $\eta = 20$

Fig. 2 Properties for the system shown in Eq. (2) under different values of η. Other parameter values are given in Eq. (13) except for (**a**) $\eta = -1000(-\infty)$, (**b**) $\eta = 10$, (**c**) $\eta = 20$. Image generated based on Ref. [89]



(a) $\alpha = 0.2$

(b) $\alpha = 0.8$

Fig. 3 Phase portraits for different values of α and K in Eq. (2). Other parameter values are given in Eq. (14) except for (**a**) $\alpha = 0.2$, (**b**) $\alpha = 0.8$. Image generated based on Ref. [89]

(p^*, z_1^*, z_2^*), and $|A| = 0$ if and only if $K = K^{tc}$. In addition, there is a $\delta > 0$ when $\mu_0 > \delta$, (p^*, z_1^*, z_2^*) is stable with any values of $K > K^{tc}$.

They found that environmental perturbations may result in the case [90]: when $\mu(p)$ is determined by the current density p, if the resource is low but increases in winter, it will reduce the stabilizing effect. However, when the portion of predators that enters dormancy is sensitive to the change of prey ($\mu(\dot{p})$), the stability can be easily observed in the ecosystem with environmental perturbations.

The impact of dormant form (resting-egg) has also been studied in the consumer-resource model [127]. In this model, the following assumptions are taken:

(1) The resting eggs are produced seasonally or due to resource deficiency;
(2) The resting eggs hatched seasonally;
(3) The gender and the quality difference of offspring are not taken into account.

Therefore, this consumer-resource model with resting eggs is shown as [128],

$$\begin{aligned} \frac{dp}{dt} &= r\left(1 - \frac{p}{K}\right)p - f(t)z_1, \\ \frac{dz_1}{dt} &= cg(t)f(t)z_1 + h(t)z_2 - d_1 z_1, \\ \frac{dz_2}{dt} &= c(1 - g(t))f(t)z_1 - h(t)z_2 - d_2 z_2, \end{aligned} \tag{17}$$

where

$$\begin{aligned} f(t) &= \frac{ap(t)}{b + p(t)}, \\ h(t) &= h_a\left\{1 + h_s \sin\left(\frac{2\pi}{T}t\right)\right\}, \\ g(t) &= g_a\left\{1 + g_s \sin\left(\frac{2\pi}{T}t\right)\right\}, \end{aligned} \tag{18}$$

where p, z_1, z_2 represent the density of phytoplankton, active and dormant zooplankton, respectively. $r, K, c, d_1, d_2, f(t)$, and $g(t)$ represent the same meaning as the parameters of previous models. $h(t)$ and $g(t)$ are the hatching rate and energy used for dormant forms

that change seasonally based on the sinusoidal function [129,130], where T and h_s (and g_s) represent the cycle and the degree of the seasonality, h_a and g_a are average hatching rate and average energy used to producing dormant forms, respectively. It has been found that $h(t)$ and $g(t)$ reach the peak in spring and autumn, respectively [66,131]. $g(t)$ not only can be determined by the seasonality, but also by resource deficiency [89] shown as,

$$\begin{aligned} g(t) &= \frac{1}{2}\left[1+\tanh\left(\frac{f(t)-x}{y}\right)\right] \\ &= \frac{1}{1+\exp\left[-2\frac{f(t)-x}{y}\right]}, \end{aligned} \tag{19}$$

which is the same as Eq. (12). Some complicated behavior will be caused by the seasonality in this Rosenzweig-MacArthur Daphnia-algae consumer-resource model [129]. Due to eutrophication in this model, large population oscillations can also cause extinction. Therefore, the environment carrying capacity K is important in this system. In addition, resting eggs have the significant impact on the system stability when they are produced by the seasonality (Eq. 18) with lower values of h_a and $g(t)$ [127]. The decline of consumer growth rate can increase the resource density in equilibrium and cause the stability of the system, so only much higher eutrophication can cause the instability of this system.

Turing instabilities [93,132,133] of homogeneous steady states have been then investigated in the following predator–prey reaction-diffusion system:

$$\begin{aligned} \frac{dp}{dt} &= d_p\Delta p + s(p) - f(p)z_1, \\ \frac{dz_1}{dt} &= d_{z_1}\Delta z_1 + k\mu(p)f(p)z_1 \\ &\quad + \alpha(p)z_2 - m(z_1)z_1, \\ \frac{dz_2}{dt} &= kv(p)f(p)z_1 - \alpha(p)z_2 - n(z_2)z_2, \end{aligned} \tag{20}$$

where k, $f(p)$ and $\mu(p)$ have the same meaning as parameters in the system shown in Eq. (2). $v(p) + \mu(p) \equiv 1$ means the energy used to different forms. $s(p)$ and $\alpha(p)$ represent the growth rate of prey and hatching rate of predator, respectively. $n(z_2)$ is the mortality rate of z_2 which is much smaller than $m(z_1)$ and both of them are increasing functions. d_p and d_{z_1} are diffusion coefficients, where d_p is also much smaller than d_{z_1}. In addition, the diffusion coefficient for z_2 is set as 0 because the dormant form (resting eggs) cannot move. When $n(z_2) = 0$ and the diffusion parts are removed, the homogeneous steady state $(\bar{p}, \bar{z}_1, \bar{z}_2)$ is the equilibrium of this system,

$$\begin{aligned} \frac{dp}{dt} &= s(p) - f(p)z_1, \\ \frac{dz_1}{dt} &= k\mu(p)f(p)z_1 + \alpha(p)z_2 - m(z_1)z_1, \\ \frac{dz_2}{dt} &= kv(p)f(p)z_1 - \alpha(p)z_2. \end{aligned} \tag{21}$$

The linearized matrix $A = (a_{ij})$ of the right side at $(\bar{p}, \bar{z}_1, \bar{z}_2)$ is defined as,

$$\begin{aligned} &a_{11} = s'(\bar{p}) - f'(\bar{p})\bar{z}_1, a_{12} = -f(\bar{p}), a_{13}=0, \\ &a_{21} = k\left(\mu'(\bar{p})f(\bar{p}) + \mu(\bar{p})f'(\bar{p})\right)\bar{z}_1 + \alpha'(\bar{p})\bar{z}_2, \\ &a_{22} = -kv(\bar{p})f(\bar{p}) - m'(\bar{z}_1)\bar{z}_1, a_{23} = \alpha(\bar{p}), \\ &a_{31} = k\left(v'(\bar{p})f(\bar{p}) + v(\bar{p})f'(\bar{p})\right)\bar{z}_1 - \alpha'(\bar{p})\bar{z}_2, \\ &a_{32} = kv(\bar{p})f(\bar{p}), a_{33} = -\alpha(\bar{p}). \end{aligned} \tag{22}$$

In general, A does not need to be stable, and the sufficient condition for the stability of A can be found in Ref. [93]. When $m'(\bar{z}_1) > 0$ and A is stable,

(1) There is no Turing instability for $(\bar{p}, \bar{z}_1, \bar{z}_2)$ in system shown in Eq. (20) if $s'(\bar{p}) - f'(\bar{p})\bar{z}_1 \le 0$, but there is Turing instability if $s'(\bar{p}) - f'(\bar{p})\bar{z}_1 > 0$.
(2) There is no oscillatory Turing instability for $(\bar{p}, \bar{z}_1, \bar{z}_2)$ in system shown in Eq. (20) if $0 < s'(\bar{p}) - f'(\bar{p})\bar{z}_1 \le \alpha'(\bar{p})$, but there is oscillatory Turing instability if $s'(\bar{p}) - f'(\bar{p})\bar{z}_1 > \alpha'(\bar{p})$.

Further proof on Turing instability can be found in Refs. [93,134,135]. The time-global unique solvability of this system is discussed in Ref. [136]. The equilibrium that is stable in the predator–prey system but does not undergo Turing instability in the corresponding predator–prey reaction-diffusion system cannot undergo Turing instability in this system with predator dormancy. Therefore, predator dormancy enhances the chance to generate Turing patterns in predator–prey systems. In addition, predator dormancy allows the equilibrium to be stable in the system, thus expanding the region where Turing instabilities of an equilibrium can be observed. Hence, predator dormancy is an enhancer rather than a generator of spatio-temporal Turing patterns in the reaction-diffusion system. Dormancy has been found as an adaptable strategy to suppress the fluctuation of population size under harsh environment, thereby preventing extinction and leading to the success of predator [137]. In addition, the system that incorporated the adaptation and defense shows that the transient complex spatio-temporal pattern is

a mixture of spatially periodic steady states or traveling/standing waves [138,139]. In the reaction-diffusion system with three coupled equations, chemotaxis can attract active predators to the region with a higher density of dormant predators [67], the boundedness of the solution [86] and the conditional convergence of the numerical method [140] have also been proven and discussed.

The fast-slow system [78,84] has also been introduced to investigate the bifurcation structure and dynamical behavior of the predator–prey system,

$$\begin{aligned} \frac{dp}{dt} &= r\left(1-\frac{p}{K}\right)p - f(p)z_1, \\ \frac{dz_1}{dt} &= k_1\mu(p)f(p)z_1 + \alpha z_2 - d_1 z_1, \\ \frac{dz_2}{dt} &= \varepsilon\left(k_2(1-\mu(p))f(p)z_1 - \alpha z_2 - d_2 z_2\right), \end{aligned} \tag{23}$$

where $\varepsilon > 0$ is a small parameter. They found that the periodic orbit will grow and gradually become a mixed-mode oscillation along the stable branch bifurcating as K increases. Many stability diagrams that can explore the chaotic phases and complex oscillatory under different parameter settings are shown in these studies [78,84].

The prey refuge, which is also a strategy to prevent extinction like dormant egg banks, has been introduced in the predator–prey system [80]. mp number of prey can be protected by the refuge, and the remaining $(1-m)p$ prey are available to the predator. This model with refuge protection is shown as,

$$\begin{aligned} \frac{dp}{dt} &= r\left(1-\frac{p}{K}\right)p - \frac{\beta(1-m)p}{1+\alpha(1-m)p}z, \\ \frac{dz}{dt} &= k\frac{\beta(1-m)p}{1+\alpha(1-m)p}z - d_z z, \end{aligned} \tag{24}$$

where $m \in [0, 1)$ is a constant, r is the intrinsic growth rate, and β/α is the maximum number of prey consumed in unit time. The predation rate in this system is also modeled by the Holling type II functional response shown in Eqs. (3) and (4). The equilibria of this system are shown as:

(1) Trivial equilibrium $(0, 0)$;
(2) Absence of predator $(K, 0)$;
(3) Positive equilibrium (p^*, z^*), where $p^* = \frac{d_z}{(k\beta - d_z\alpha)(1-m)}$ and $z^* = \frac{\alpha k}{K}\frac{K(k\beta-d_z\alpha)(1-m)-d_z}{((k\beta-d_z\alpha)(1-m))^2}$. The positive equilibrium only exists when $k\beta > d_z\alpha$ and $0 \le m < 1 - \frac{d_z}{K(k\beta-d_z\alpha)}$.

Let $r = 10, K = 100, \alpha = 0.02, \beta = 0.6, d_z = 0.09, k = 0.02$ in appropriate units [80]. The stability of the equilibrium in this system is shown in Fig. 4. When $m = 0.25$ (smaller than the bifurcation point 0.32), the positive equilibrium in this system is unstable and there is a limit cycle to show the global stability (Fig. 4a and b). When m is larger than the bifurcation point, there will be a positive equilibrium in the system, thus allowing the system to be stable. The population curves for $m = 0.5$ is shown in Fig. 4c, and the phase portrait for $m = 0.5$ with different initial population sizes is shown Fig. 4d with a global attractor (17.65, 32.3). Therefore, refuges have an important impact on pest control, but more refuges will also cause higher density of prey, such as the "hotspots" of high spider mite density in orchard [141]. The existence, stability, and persistent criteria of this system have been studied in Ref. [80].

4 Competition between dormitive and perennially active species

In Sect. 3, there is only the dormitive predator in predator–prey systems, such that prey is only consumed by the dormitive predator without competition. However, the advantage of dormancy cannot be demonstrated through comparative experiments when there is only one type of species. Therefore, there will be two compartments, that is, active and dormitive species, with different features to compete in predator–prey systems shown in this section.

4.1 Predator dormancy

The predator–prey system that contains both the active and dormitive predator [94] adapted from the Rosenzweig and MacArthur [27,142] is introduced to show their competition:

$$\begin{aligned} \frac{dp}{dt} &= r\left(1-\frac{p}{K}\right)p - f_y(p)y - f_z(p)z_1, \\ \frac{dy}{dt} &= k_y f_y(p)y - d_y y, \\ \frac{dz_1}{dt} &= k_{z_1}\mu(p)f_z(p)z_1 + \alpha z_2 - d_{z_1}z_1, \\ \frac{dz_2}{dt} &= k_{z_2}(1-\mu(p))f_z(p)z_1 - \alpha z_2 - d_{z_2}z_2, \end{aligned} \tag{25}$$

where p, y, z_1, and z_2 represent the prey, active predator, active form and dormant form of dormitive preda-

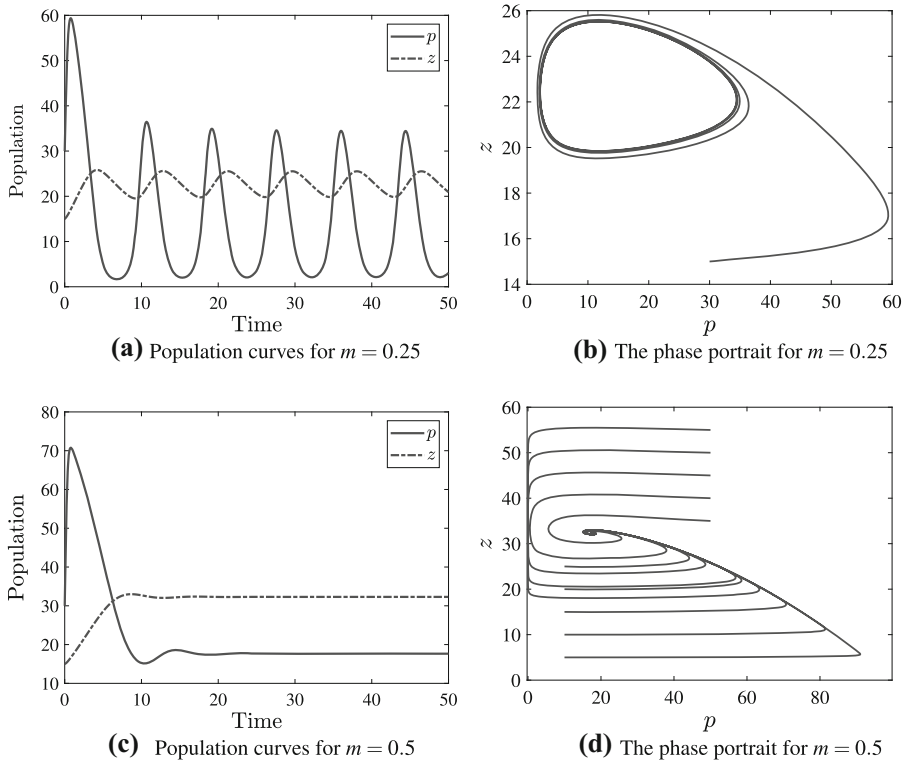


Fig. 4 Population curves and phase portraits for the dynamics with prey refuge (Eq. 24) under different values of m. (**a**) Population curves for $m = 0.25$, (**b**) Phase portrait for $m = 0.25$, (**c**) Population curves for $m = 0.5$, (**d**) Phase portrait for $m = 0.5$ with a global attractor (17.65, 32.3). Image generated based on Ref. [80]

tor, respectively. The predation rate and the portion of energy used for active forms are specified by

$$f_q(p) = \frac{c_q p}{1 + c_q h_q p}, q \in \{y, z\},$$
$$\mu(p) = \left[1 + \exp\left(\frac{-(p-\eta)}{\sigma}\right)\right]^{-1}. \tag{26}$$

The predation rates $f_y(p)$ and $f_z(p)$ have different parameter settings so that two types of predator have different abilities. The diagrams of $f(p)$ and $\mu(p)$ with some important properties are shown in Fig. 5.

For this system with two types of predators, apart from the trivial equilibrium $x_0^* = (0, 0, 0, 0)$, it can be stable with only one predator:

(1) Dormitive-only equilibrium $x_z^* = (p^*, 0, z_1^*, z_2^*)$. After setting $p', z'_1, z'_2 = 0$, we obtain

$$\begin{aligned} d_{z_1} &= k_{z_1}\mu(p^*) f_z(p^*) \\ &\quad + \bar{k}_{z_2}(1 - \mu(p^*)) f_z(p^*), \\ z_1^* &= \frac{r(1 - p^*/K)p^*}{f_z(p^*)}, \\ z_2^* &= \frac{\bar{k}_{z_2}}{\alpha}(1 - \mu(p^*)) f_z(p^*) z_1^*, \end{aligned} \tag{27}$$

where $\bar{k}_{z_2} = \alpha k_{z_2}/(\alpha + d_{z_2})$. y can be seen as z_1 when $\alpha = 0$ and $\mu(p) = 1$ in this situation.

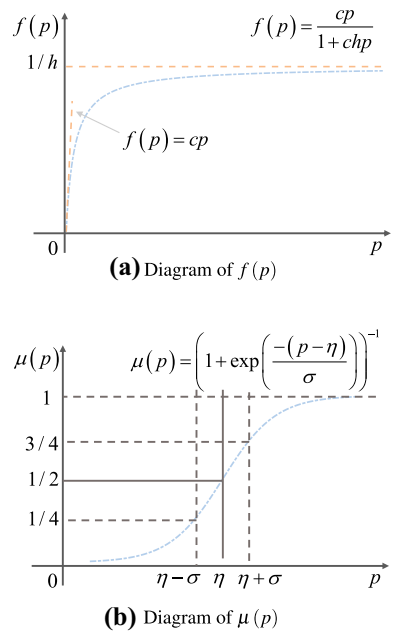


Fig. 5 Graphs of $f(p)$ and $\mu(p)$

(2) Active-only equilibrium $x_y^* = (p^*, y^*, 0, 0)$. Setting $p', y' = 0$ yields

Table 1 The descriptions, units, and values of parameters in the predator–prey system shown in Eq. (25)

Parameter	Description	Value	Units
p	Density of prey	Variable	mg L^{-1}
y	Density of perennially active predator	Variable	mg L^{-1}
z_1	Density of active form of dormitive predator	Variable	mg L^{-1}
z_2	Density of dormant form of dormitive predator	Variable	mg L^{-1}
r	Prey growth rate	0.5	Day^{-1}
K	Environmental carrying capacity	Variable	mg L^{-1}
k_y	Growth efficiency of active predator y	0.5	–
k_{z_1}, k_{z_2}	Growth efficiencies of dormitive predator z	0.5, 0.25	–
d_y	Mortality rate of active predator y	0.25	Day^{-1}
d_{z_1}, d_{z_2}	Mortality rates of dormitive predator z	0.25, 0.05	Day^{-1}
α	Hatching rate	0.05	Day^{-1}
c_y, c_z	Foraging efficiencies	0.4	Day^{-1} mg^{-1} L
h_y, h_z	Handling times	0.75	Day
η	Dormancy switching threshold	2.5	mg L^{-1}
σ	Dormancy switching width	1	mg L^{-1}
$f_y(p), f_z(p)$	Predation rate function	Function	Day^{-1}
$\mu(p)$	Dormancy switching function	Function	–

The values are obtained from previous plankton ecosystems [84,125,143]. Table reproduced from Ref. [94] with explicit permission

$$d_y = k_y f_y\left(p^*\right),$$
$$y^* = \frac{r\left(1 - p^*/K\right)p^*}{f_y\left(p^*\right)}, \tag{28}$$

which can be obtained from Eq. (27) by setting $\alpha = 0$ and $\mu\left(p\right) = 1$.

However, there is no equilibrium signifying the coexistence of the active and dormitive predators [94]. The argument for this is the following: In general, the abilities of active forms are stronger than dormant forms, including the predation rates $f_y(p) \geq f_z(p)$, growth efficiencies $k_y \geq k_{z_1} > k_{z_2}$, and mortality rates $d_y \leq d_{z_1}$. By setting $y', z'_1, z'_2 = 0$ for the equilibrium $(\bar{p}, \bar{y}, \bar{z}_1, \bar{z}_2)$ with $\bar{y}, \bar{z}_1, \bar{z}_2 > 0$, it is easily to obtain

$$\begin{aligned} d_{z_1} &= k_{z_1}\mu\left(\bar{p}\right) f_z\left(\bar{p}\right) + \bar{k}_{z_2}\left(1 - \mu\left(\bar{p}\right)\right) f_z\left(\bar{p}\right), \\ d_y &= k_y f_y\left(\bar{p}\right). \end{aligned} \tag{29}$$

With the setting of the mortality rates $d_y \leq d_{z_1}$, the inequality can be shown as $\left(k_{z_1} - \bar{k}_{z_2}\right)\mu\left(\bar{p}\right) f_z\left(\bar{p}\right) \geq k_y f_y\left(\bar{p}\right) - \bar{k}_{z_2} f_z\left(\bar{p}\right)$. Due to the predation rate $f_y(p) \geq f_z(p)$, this inequality $\mu\left(\bar{p}\right) \geq \left(k_y - \bar{k}_{z_2}\right) / \left(k_{z_1} - \bar{k}_{z_2}\right)$ can be obtained. Furthermore, due to $\mu\left(\bar{p}\right) < 1$ and $k_y \geq k_{z_1} \geq k_{z_2} \geq \bar{k}_{z_2}$, the inequality does not hold and the equilibrium $(\bar{p}, \bar{y}, \bar{z}_1, \bar{z}_2)$ with $\bar{y}, \bar{z}_1, \bar{z}_2 > 0$ does not exist.

The descriptions, units, and values of these parameters in this model can be found in Table 1. A small number of active predator, y can easily make the system unstable from the dormitive-only equilibrium x_z^*, and the system eventually reaches the active-only equilibrium x_y^* (Fig. 6). The initial population size is $x_z^* = \left(p^*, 0, z_1^*, z_2^*\right)$, and y gradually increases to $z_1^*/4$ at time $t = 0$ (mutation) which follows the condition $y + z_1 = z_1^*$. Here, the active predator and the active form of the dormitive predator have the same features $k_y = k_{z_1}, d_y = d_{z_1}, f_y = f_z$ (Table 1), but extra resources are needed to produce dormant progeny z_2 for the dormitive predator. From Fig. 6, the predator dormancy is unstable without population cycles, and the system can easily reach the active-only equilibrium x_y^* with a small number of initial size y. Therefore, the dormitive predator z cannot stop the emergence of the active predator y. On the contrary, the active-only equilibrium x_y^* is stable with the change of z_1 and z_2, which can prevent extinction of populations.

Both the active form and the dormant form of the dormitive predator cannot outcompete the active preda-

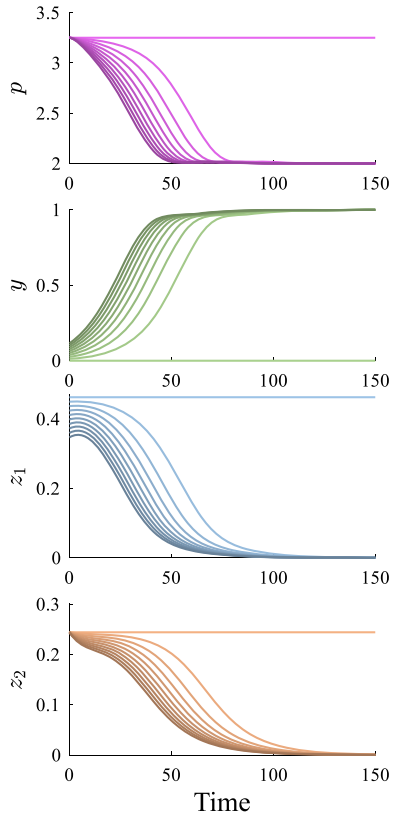


Fig. 6 The instability of the dormitive-only equilibrium x_z^*. The parameter values are from Table 1 except for $K = 4$. The initial population size is x_z^*, and y increases from 0 to $z_1^*/4$ under the condition $y + z_1 = z_1^*$. The darker the line, the larger initial number of y. Image reproduced from Ref. [94] with explicit permission

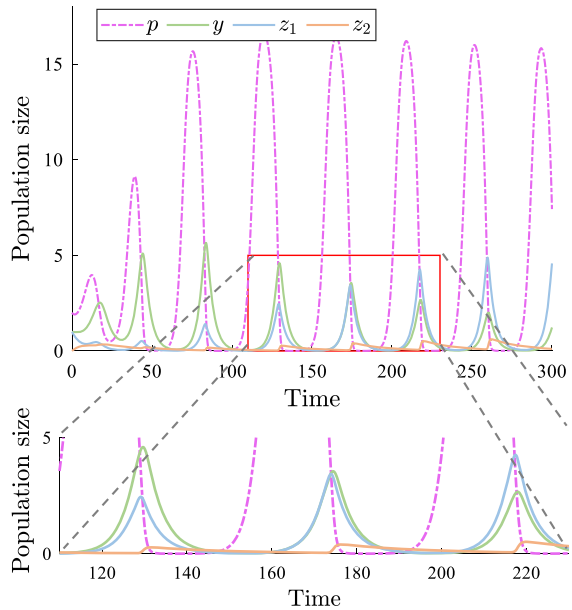


Fig. 7 The competition between the active predator and dormitive predator with large amplitude population cycles. The parameter values are from Table 1 except for $K = 18$, $d_{z_2} = 0.025$, and $\alpha = 0.025$. The active form recovers faster than the dormant from in the predation, and the dormant form is the reservoir under the shortage of prey (food and resource). The dormitive predator can outcompete the active predator by switching between losing strategies. Image reproduced from Ref. [94] with explicit permission

tor individually: (1) active forms need to spend extra resources producing the dormant progeny, and (2) dormant forms need more time and resources than active forms because they cannot produce progeny on its own and need to consume energy to produce resting eggs [78,79], like zooplankton. However, the dormitive predator can win the competition by alternating between two losing strategies through the threshold-induced switching function shown in Eq. (26). From Fig. 7, the dormitive predator is still the disadvantaged predator at the beginning. The density of z_1 gradually increases and outcompetes y around $t = 220$ shown in the enlarged segment. Interested readers can refer to Ref. [94] for a more detailed discussion pertaining to the stability of the system in the long term. Hence, dormitive predator with inferior traits can achieve a higher density than perennial active predator under conditions of extensive prey density fluctuation due to overpredation. This counter-intuitive winning result for the dormitive predator caused by alternating between the active and dormant forms (two losing strategies) can be explained by the game-theoretic Parrondo's paradox [144–150], and suggests that dormancy may be a natural evolutionary response to the self-destructive tendencies of overpredation [94].

4.2 Prey dormancy

The prey can also enter dormancy to prevent extinction under harsh environmental conditions (high density of predators) [112]. Therefore, the predator–prey system that has two types of prey with different features under predation is shown as,

$$\frac{dp}{dt} = r_p\left(1 - \frac{p + y_1}{K}\right)p - f_p(p)z,$$

$$\frac{dy_1}{dt} = r_y\left(1 - \frac{p+y_1}{K}\right)\mu(z)y_1 + \alpha y_2 - f_{y_1}(y_1)z,$$

Table 2 The descriptions, units, and values of parameters in the predator–prey system shown in Eq. (30)

Parameter	Description	Value	Units
p	Density of perennially active prey	2	mg L^{-1}
y_1	Density of active form of dormitive prey	2	mg L^{-1}
y_2	Density of dormant form of dormitive prey	0	mg L^{-1}
z	Density of predator	1	mg L^{-1}
r_p, r_y	Prey growth rates	0.55, 0.5	Day^{-1}
K	Environmental carrying capacity	11 (Variable)	mg L^{-1}
k_p, k_{y_1}	Growth efficiencies of predator from active form	0.5	–
k_{y_2}	Growth efficiency of predator from dormant form	0.25	–
d_z	Mortality rate of predator	0.2	Day^{-1}
α	Termination rate of dormancy	0.05	Day^{-1}
c_p, c_{y_1}, c_{y_2}	Foraging efficiencies	1, 1, 0.4	Day^{-1} mg^{-1} L
h_p, h_{y_1}, h_{y_2}	Handling times	0.5	Day
η	Dormancy switching threshold	1	mg L^{-1}
σ	Dormancy switching width	0.1	mg L^{-1}
χ	Lower bound	0.2	mg L^{-1}
φ	Range	0.75	mg L^{-1}
$f_p(p)$, $f_{y_1}(y_1)$, $f_{y_2}(y_2)$	Predation rate function	Function	Day^{-1}
$\mu(p)$	Dormancy switching function	Function	–

Table reproduced from Ref. [112] with explicit permission

$$\begin{aligned} \frac{dy_2}{dt} &= r_y\left(1-\frac{p+y_1}{K}\right)(1-\mu(z))\,y_1 \\ &\quad -\alpha y_2 - f_{y_2}(y_2)z, \\ \frac{dz}{dt} &= k_p f_p(p)z + k_{y_1} f_{y_1}(y_1)z + k_{y_2} f_{y_2}(y_2)z - d_z z, \end{aligned} \tag{30}$$

where p is the active prey, y_1 and y_2 are active form and dormant form of the dormitive prey, and z is the predator. The predation rate ($f_p(p)$, $f_{y_1}(y_1)$, and $f_{y_2}(y_2)$) and the portion of energy used for active forms $\mu(z)$ are similar to Eq. (26) and shown as,

$$\begin{aligned} f_q(q) &= \frac{c_q q}{1+c_q h_q q}, q \in \{p, y_1, y_2\}, \\ \mu(z) &= \chi + \varphi\left[1+\exp\left(\frac{z-\eta}{\sigma}\right)\right]^{-1}, \end{aligned} \tag{31}$$

where χ and φ are the lower bound and range of function $\mu(z)$. The descriptions, units, and values of these parameters in this model can be found in Table 2 in detail.

The dormitive prey is also the disadvantaged species in the competition because (1) the active form y_1 has a lower growth rate and needs to spend extra resource producing dormant progeny; (2) the dormant form y_2 has a lower growth rate and it cannot increase on its own. The losing results for the dormitive prey y are shown in Fig. 8a and b, where there is only y_1 or y_2 in the losing strategy I and strategy II, respectively. Both strategies lead to the extinction of y, thus causing the losing outcome. However, alternating between these two losing strategies (the setting of $\mu(p)$) allows the disadvantaged prey y to win the competition (Fig. 8c). The first time that the peak density of y_1 is higher than the peak density of p is called the reversal time $\bar{T}$. $\bar{T}$ will change rapidly when K is low but almost remain unchanged with high value of K (Fig. 8f). In detail, the reversal time $\bar{T}$ equal 195.8, 142.5, and 87.3 when the environment carrying capacity K are 12, 15, and 18, respectively (Fig. 8c–e). Therefore, the carrying capacity K has a great impact on the reversal time in the competition.

The parameter spaces that determine the result of competition have also been studied, and the population sizes ($y_1 - p$) are obtained by the highest density of p and y_1 after the competition is stable ($t \in [2800, 3000]$). Parameters are compared in pairs because of the impracticality of all-vs-all comparison. p has higher density when K is low no matter how α changes, and y_1 is the advantaged prey when K

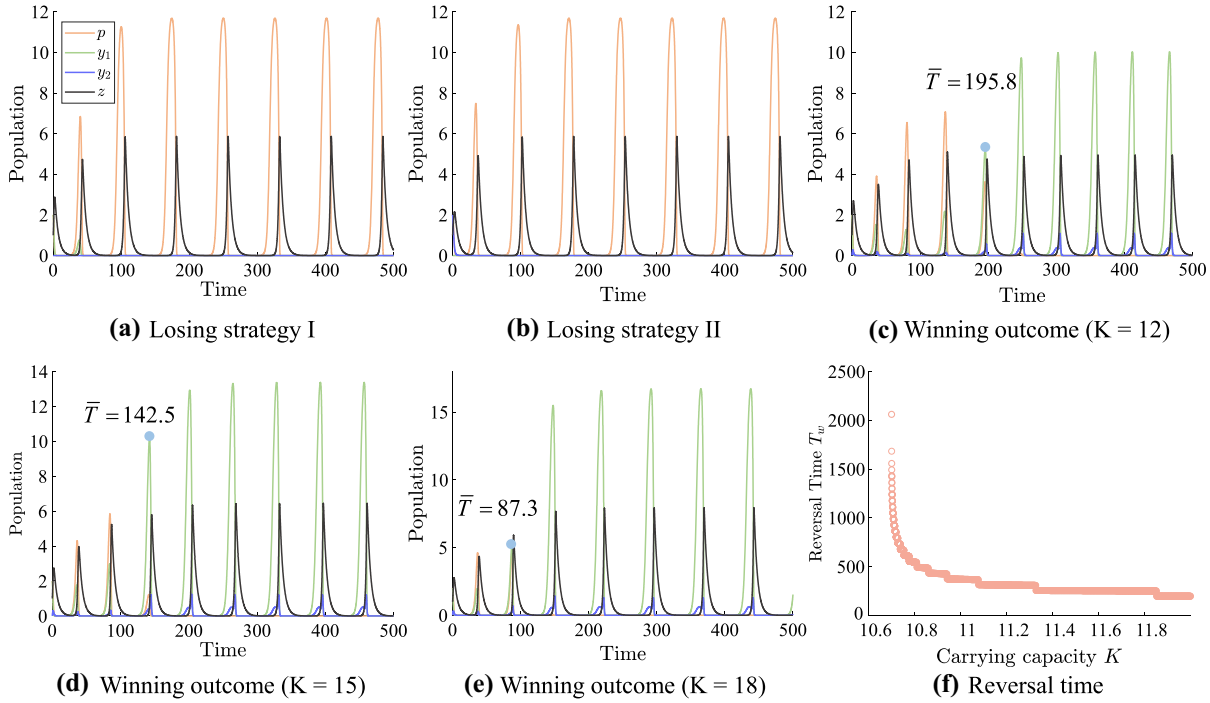


Fig. 8 The competition between the active prey and dormitive prey under predation. The parameter values are from Table 2 unless stated otherwise. (**a**) Only the active form competes with the perennially active prey ($\alpha = 1, \mu(z) = 1, K = 12$). (**b**) Only the dormant form competes with the perennially active prey ($\alpha = 0, \mu(z) = 0, K = 12$, and the initial population size is [2, 0, 2, 1]). (**c**) Both y_1 and y_2 compete with the perennially active prey ($K = 12$). (**d**, **e**) The competition with higher value of K ($K = 15, K = 18$) can cause higher density and shorter reversal time. (**f**) The relationship between $\bar{T}$ and K. Image adapted from Ref. [112]

becomes larger with medium values of α (Fig. 9a). There will be more predators with lower value of mortality rate d_z, thus easily causing the extinction of p because p cannot enter dormancy to prevent the large amplitude cycle (Fig. 9b). The impact of parameters in $\mu(z)$ on the competition is shown in Fig. 9c and d. The dormitive prey can only be the advantaged population when $\chi + \varphi$ is close to 1 and χ is small (Fig. 9c). In addition, y_1 can only have higher density than p with small value of σ (Fig. 9d). This is because only the sharp switching between y_1 and y_2 under harsh environmental conditions allows the dormitive prey to have a strong fitness and win the competition. The predator growth efficiency from the dormant form k_{y_2} hardly affects the competition, and the dormitive prey can obtain higher fitness with larger predator growth efficiencies k_p and k_{y_1} due to the predation of z (Fig. 9e). Similarly, y_1 becomes the advantaged prey with higher values of c_p and c_{y_1} (Fig. 9f). Interested readers can refer to Ref. [112] for a more detailed discussion pertaining to dominance in the long term. Under high density of predators, prey needs to avoid being foraged, thus dormancy becomes the beneficial strategy to the prey because dormant form has lower consumption and breeding, as well as in the probability of being discovered. In addition, active form will promote the growth of prey when the density of predators is low. Therefore, dormancy is beneficial for organisms under large-amplitude environmental fluctuation, especially under the high density of predators. Dormancy becomes an effective strategy to help the disadvantaged prey to win the competition in the predator–prey system. This is analogous to Parrondo's paradox: neither the active nor dormitive forms of the dormant prey can achieve a higher density than a perennially active predator individually, but a winning outcome can be achieved by alternating between these two losing strategies.

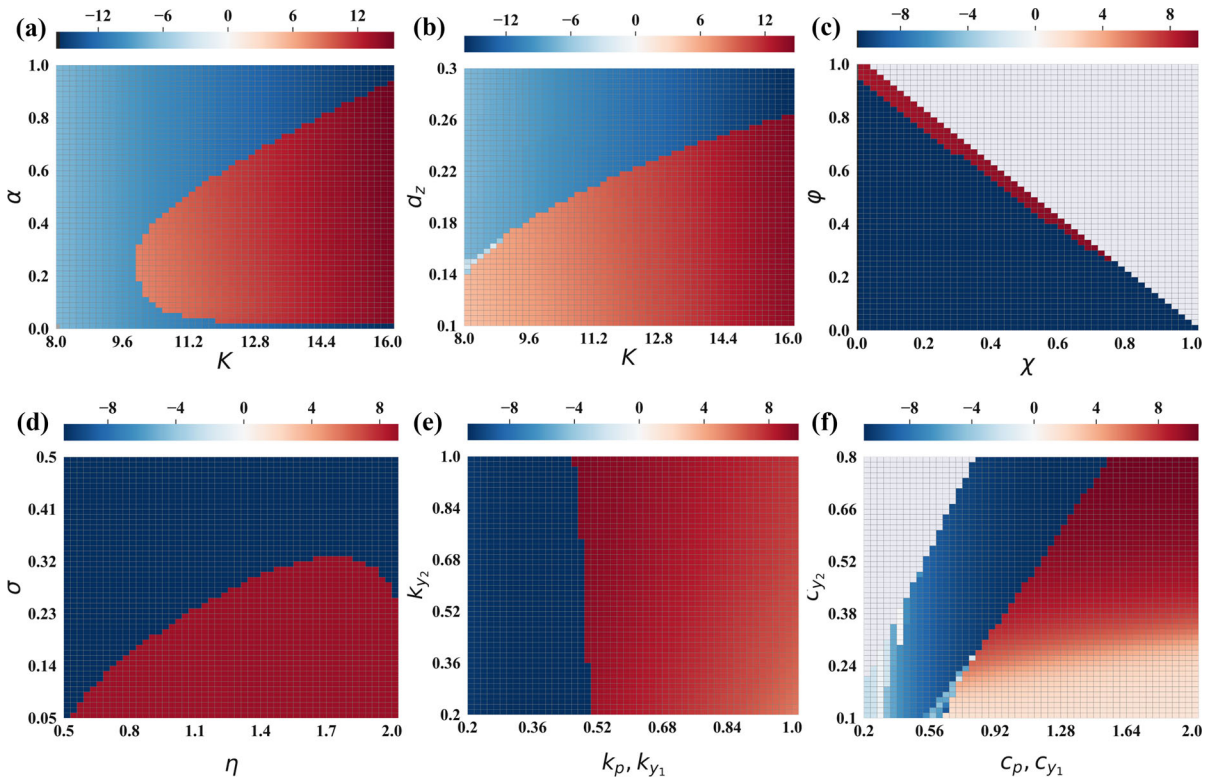


Fig. 9 Parameter spaces related to the competition between y_1 **and** p in Eq. (30). The parameter values are from Table 2 unless stated otherwise. (**a**, **b**) Joint impact of K, α, and d_z on the competition. (**c**, **d**) Joint impact of parameters in $\mu(z)$ on the competition ($0 \leq \chi \leq \chi + \varphi \leq 1$ in (**c**)). (**e**, **f**) The impact of predator features on the competition ($c_{y_2} \leq c_p$ and $c_{y_2} \leq c_{y_1}$ in (f)). Image adapted from Ref. [112]

5 Conclusion and future outlook

The predator–prey system describes the interaction between species, thus uncovering the evolution and competition of populations in the environment. Dormancy is a common and important form of stress response under environmental fluctuation. In this review paper, the classical predator–prey system (Lotka–Volterra model) is introduced in Sect. 2 first. A variety of predator–prey systems with dormancy are discussed in Sect. 3, including important properties, system stability analysis, and impacts of parameters. Dormitive predator (prey) is then introduced in the system to compare with the perennially active predator (prey) to show the benefit of dormancy under harsh environmental conditions in Sect. 4. In addition, the dormant form and active form of dormitive predator (prey) are two losing strategies in the competition, but these losing strategies can be combined to achieve a winning outcome through Parrondo's paradox. Therefore, dormancy can be seen as an effective response to self-destructive overpredation under large-amplitude fluctuation, which is significant in predator–prey systems.

Future direction in predator–prey systems are also discussed:

(1) Modeling dormancy in predator–prey systems: Existing researches are concentrated on using the sigmoid switching function to model the induction of dormancy $\mu(p)$. Are there other functions that can be more consistent with biological activities to describe the behavior? There are still many open problems related to dormancy in predator–prey systems.
(2) Other biological systems with dormancy: Beyond predator–prey systems, species in other biological systems can also enter dormancy to prevent extinction under large-amplitude population fluctuations, like host-parasite, lysis-lysogeny,- and

virus-immune systems. In addition, Parrondo's paradox appears to be a useful mathematical tool that can be used in more biological systems to explain how the disadvantaged species can win the competition by alternation. Therefore, dormancy should be further studied in other biological systems.

(3) Species traits in dynamical systems: In general, changes in the species and the environment will have an impact on each other. However, almost all traits of species in predator–prey systems are regarded as static values, such as the growth rate, mortality rate and predation rate. These traits of species may be considered as dynamic values (as the system changes) in future studies.

(4) Validation through laboratory experiments: Current studies mainly uncover the significance of dormancy in predator–prey systems through simulations by the aid of real-world data, and more experiments are needed in the laboratory to validate some of these novel results arising from the simulations.

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Data availability All data generated or analyzed during this study are included in this article.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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